

ARGHA PRATIM SAHA

arghapratimsaha00@gmail.com | Dhaka, Bangladesh
[LinkedIn](#) | [Google Scholar](#) | [GitHub](#) | [Portfolio](#)

Education

BSc in Computer Science and Engineering [Feb 2020 – July 2025]
Shahjalal University of Science and Technology (SUST), Sylhet
CGPA: 3.71 / 4.00
Thesis: *CyQured: Design, Development, and Empirical Evaluation of a Tabletop Game for Personal Cybersecurity Education*
Author: **Argha Pratim Saha**, Utsho Das
Supervisor: Md Masum, Dr. Farida Chowdhury, Dr. Md Sadek Ferdous

Research Interests

Human-Computer Interaction (HCI) | Usable Security and Privacy | Human-Centered Cybersecurity
| Security Education and Mental Models | Human-AI Interaction

Publications

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1. CyQured: Design, Development, and Empirical Evaluation of a Tabletop Game for Personal Cybersecurity Education
USENIX Symposium on Usable Privacy and Security (SOUPS), 2026. *Accepted*
Utsho Das, Argha Pratim Saha*, Md Sadek Ferdous, Md Masum, Farida Chowdhury*
*Equal contribution.
 2. A Deep Learning Approach to Automate Classification of Arsenic-Affected Skin using EfficientNet-B1
27th International Conference on Computer and Information Technology (ICCIT). [Dec 2024]
Argha Pratim Saha, Utsho Das, Shahidur Rahman
 3. Project PRIMUS at EHRSQL 2024: Text-to-SQL Generation using Large Language Models for EHR Analysis
6th Clinical Natural Language Processing Workshop, NAACL 2024, Mexico City. [June 2024]
Sourav Bhowmik Joy, Rohan Ahmed, Argha Pratim Saha, Partha Sarothi Bhowmik, Utsho Das, Minhaj Ahmed

Research Experience

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- **Research Assistant (part-time), BRAC University, Dhaka** [Jan 2026 – Present]
Supervisor: Dr. Farida Chowdhury
 - Conducting research on how young adults in Bangladesh understand and respond to phishing and smishing, with a focus on designing interventions that do not assume prior security knowledge.
 - Mentoring undergraduate thesis groups on problem formulation, literature review, study design, and writing.
 - **Undergraduate Thesis Research, SUST** [2024 – 2025]
Supervisor: Dr. Md. Masum
 - CyQured: Designed a tabletop game that maps the STRIDE threat framework onto smart-home scenarios, enabling people without formal security training to reason about concrete cybersecurity threats.
 - Conducted a preliminary calibration study for the knowledge assessment, followed by a mixed-methods evaluation with 50 undergraduate participants.
 - Analysed pre- and post-test outcomes using linear mixed-effects models, finding significant gains in threat identification alongside positive usability outcomes.
 - **Research Project, SUST** [2024]
 - Investigated deep learning approaches for automated classification of arsenic-affected skin images, comparing multiple lightweight models with consideration for deployment in resource-constrained and remote settings.
 - Conducted model training, evaluation, and comparative analysis, resulting in a publication at

Industry Experience

Associate Software Engineer, ShellBeeHaken Ltd., Dhaka [Oct 2025 – Present]

- **KriyaKarak | Visit:** Built core front-end modules for an open marketplace for local creative services, including service-provider profiles and reservation workflows; conducted end-to-end usability testing and resolved critical functional and usability issues.
- **AI R&D:** Designed and developed a voice-first conversational AI agent for Bangla-speaking agricultural entrepreneurs and farmers, supporting real-time conversational interaction, interruption handling, and multimodal text, audio, and video communication.
- Built the real-time communication infrastructure using WebSockets and implemented budget- and goal-based planning to support personalized interactions and task-oriented assistance.
- Developed a retrieval-augmented generation (RAG) pipeline grounded in a domain-specific knowledge base, incorporating safeguards against adversarial and unsafe inputs. Explored architectures for extending the system to low-connectivity, phone-based access for users in remote areas.

Problem Solving & Competitive Programming

Solved over 1,000 algorithmic problems across various automated evaluation judges (including LeetCode, Codeforces, LightOJ, HackerRank, HackerEarth, and VJudge).

- **LeetCode:** [bob_dylan](#) | *Contest Rating: 1900+ (5 Earned Badges, 360+ Problems Solved).*
- **Codeforces:** [fugitive_stranger](#) | *Max Rating: 1419 (Specialist).*
- ICPC Bangladesh Preliminary 2021.

Selected Projects

- **AutoScriber** [[GitHub](#)] [[Website](#)] – A transcription pipeline for qualitative HCI researchers working with Bengali interviews, supporting audio preprocessing, silence removal, and transcription through both local open-source models and cloud APIs with explicit privacy trade-offs. Designed to help researchers balance transcription accuracy with the privacy and ethical requirements of handling sensitive interview data. [*TypeScript, Node.js, CSS*].
- **git-user** [[GitHub](#)] – A cross-platform Go tool for switching between multiple Git identities and encrypted SSH keys, with passphrase-based encryption for stored credentials. Published on npm as [git-userhub](#), with over 12,000 downloads in three months. [*Go, JavaScript*].

Technical Skills

- **Languages:** C/C++, Python, Go, JavaScript, TypeScript, Java
- **Web and systems:** React, Next.js, Node.js, Express, FastAPI, WebSockets, MySQL, MongoDB, Qdrant
- **Tools:** Git, L^AT_EX, Python statistical libraries

Leadership & Service

Human-Centered Computing Society (HCCS) [Jan 2026 – Present]

BRAC University, Dhaka

Student Executive Team Co-Coordinator [[website](#)]

- Coordinate research-focused seminars and activities, facilitate collaborative research ideation and team formation, and contribute to developing research guidelines.

References

Dr. Farida Chowdhury

Professor, Department of Computer Science and Engineering

BRAC University, Dhaka

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Thesis supervisor and current research supervisor.

Dr. Md. Sadek Ferdous

Professor, Department of Computer Science and Engineering

BRAC University, Dhaka

sadek.ferdous@bracu.ac.bd

Undergraduate thesis supervisor.